Experiment No: 07

Experiment Name: Single Side Band (SSB) Modulation and Demodulation using MATLAB.

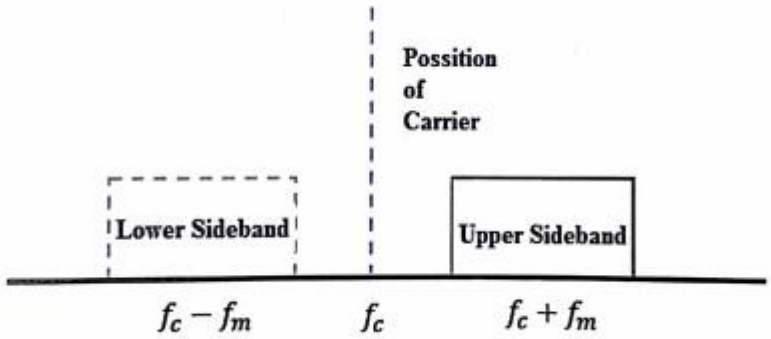
Objective:
This experiment deals with the basics of Single Side Band Suppressed Carrier (SSB-SC) modulation, and demodulation techniques for basic communication engineering. The student will learn the basic concepts of SSB modulation and through the theoretical knowledge of course, upon successful completion of the experiment the student will:

1. Understand SSB modulation and the difference between SSB and DSB modulation
2. Understand the concept of phase shift technique
3. Estimation of sideband suppression and process of SSB

Equipment's/ Apparatus Required:

1. Computer with Matlab

Theory:
Single sideband (SSB) modulation might be a sort of modulation used to send data, including audio and analog signals. Amplitude modulation produces two sidebands and a carrier. However, the SSB theory filters the frequency spectrum of the original signal. SSB modulation allows the transmission of only one sideband, either the upper or lower sideband.



The basic concept of SSB is to suppress the carrier and one of the sidebands. The suppressed technique is used in three steps to get the left-hand of the signal by using phase shift method. The first step is to detect the message signal from the source and modulate the carrier. Third step is finding the signal through demodulation and filtering.

To understand this technique, we took signal and gave it to two signals that apply the same shift to carrier (high frequency), but with 90° phase shift between these two signals. These two signals are then combined in such a way that the signal without the phase shift and the carrier without shift will cancel each other and the signal with phase shift will remain and appear without the carrier. This is the basic concept of phase shift technique used to demodulate the signal.

The advantage of using SSB modulation is that the signal is suppressed on left side and this allows the user to use less band for transmission. Using this technique we allow to demodulate the signal from our previous concept of demodulation and then demodulate the signal.

We can use the technique of carrier multiplication as we can say product modulator. An SSB can be demodulated with a product demodulator. But a product demodulator is not seen as the SSB demodulator in the specific sense. Some SSB demodulator can distinguish between a second sideband and an upper sideband. This experiment even investigates the phasing in the demodulator. It would be helpful, though not stated, that the Lab sheet entitled SSB generation has been completed.

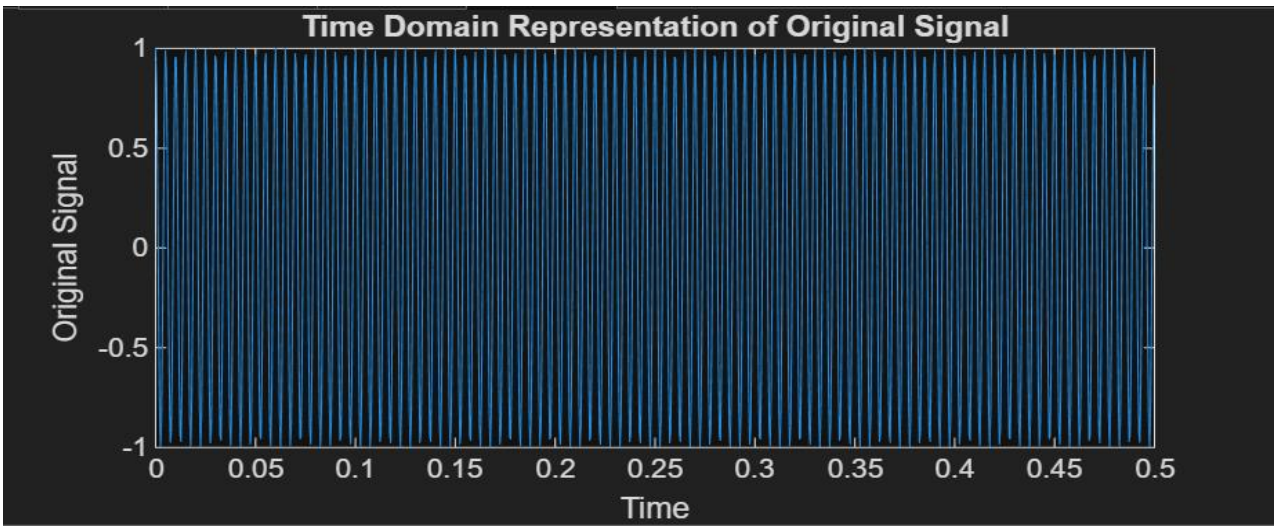
In order to receive the carrier phase shift signal we multiply the modulated signal with carrier signal having without phase shift. As a result we get the phase shift signal we multiply the modulated signal with carrier signal with phase shift. After that a signal with low pass filter is used to get the desired signal as a result.

MATLAB Code

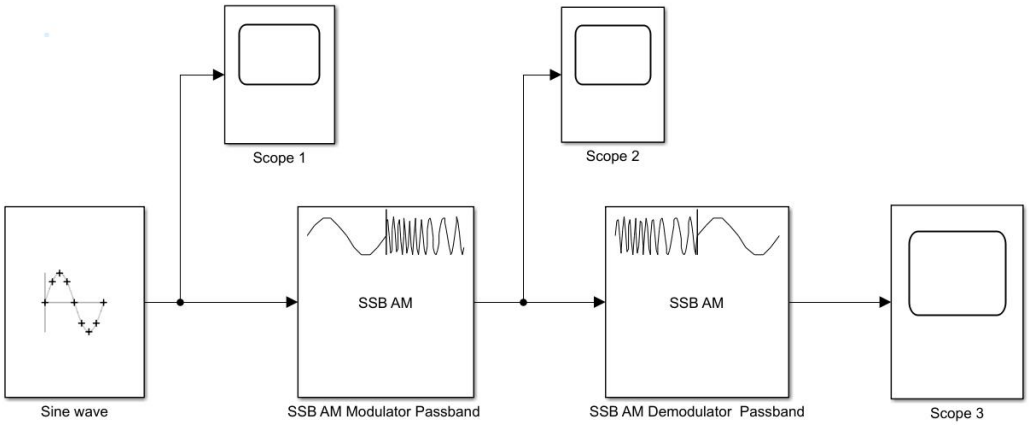
```
%E-241440
clc
clear all
N = 1024;
fs = 2048;
ts = 1/fs;
%t=(0:ts:1);
t=(0:N-1)/fs;
fc = 600; %Carrier frequency !! Limit fc<800 to avoid freqdomain aliasing
fm1 = 200;
Em1 = 1;
m= Em1*cos(2*pi*fm1*t) %Message
mh =Em1*cos((2*pi*fm1*t)-pi/2); %Hilbert transform of the message signal
sbu=m.*2.*cos(2*pi*fc*t)-mh.*2.*sin(2*pi*fc*t) %Expression for USB SSB
sbl=m.*2.*cos(2*pi*fc*t)+mh.*2.*sin(2*pi*fc*t) %Expression for LSB SSB
SBU = 2/N*abs(fft(sbu)); %Fourier Transform of USB SSB
SBL=2/N*abs(fft(sbl));%Fourier Transform of LSB SSB
freq=fs*(0:N/2)/N;
close all
figure(2)
subplot(221);
plot(10*t(1:200),sbu(1:200),'r'); %Time Domain Plot of USB SSB
title('Time Domain Representation === USB');
xlabel('Time');
ylabel ('Modulated Signal');
subplot(222);
plot(10*t(1:200),sbl(1:200),'b'); %Time Domain Plot of LSB SSB
title('Time Domain Representation===LSB');
xlabel('Time');
ylabel('Modulated Signal');
subplot(223);
plot(freq,SBL(1:N/2+1));
title('Frequency Domain Representation');
```

```
 title('Frequency Domain Representation');
 xlabel('Frequency(Hz)');
 ylabel('Spectral Magnitude');
 legend('USB');
 subplot(224)
 plot(freq,SBL(1:N/2+1)); %Frequency domain plot
 title('Frequency Domain Representation');
 xlabel('Frequency(Hz)');
 ylabel('Spectral Magnitude');
 legend('LSB');
 figure(4)
 plot(freq,SBU(1:N/2+1),freq,SBL(1:N/2+1));
 title('Frequency Domain Representation');
 xlabel('Frequency(Hz)');
 ylabel('Spectral Magnitude');
 legend('USB','LSB');
% Demodulation:
md=sbu.*cos(2*pi*fc*t);
[b,a]=butter(2,0.1);
mf=filter(b,a,md);
figure(3)
plot(t,mf)
title('Demodulated Signal');
xlabel('Time');
ylabel('Demodulated Signal');
figure(1);
plot(t,m);
title('Time Domain Representation of Original Signal');
xlabel('Time');
ylabel('Original Signal');
```

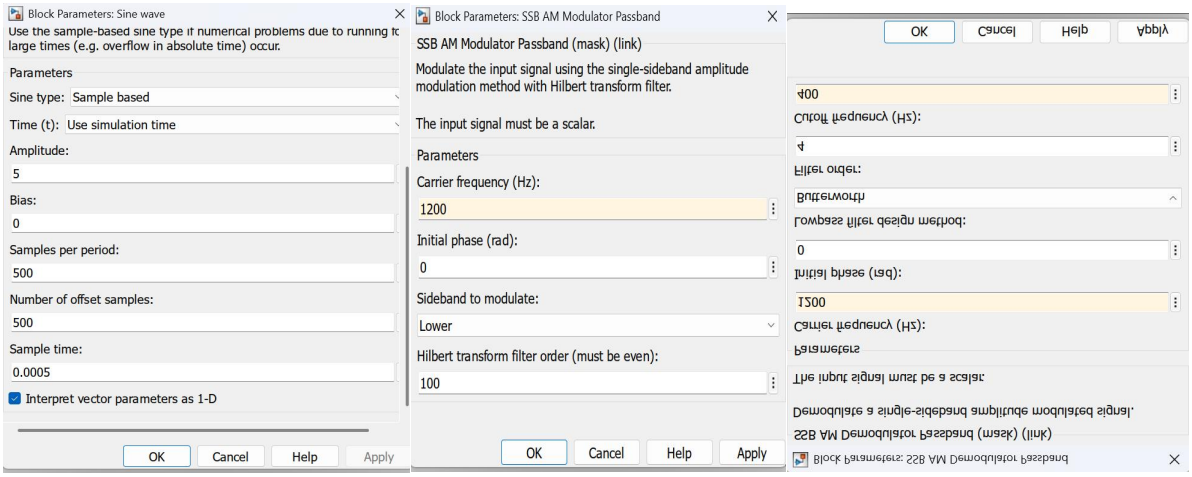
Output



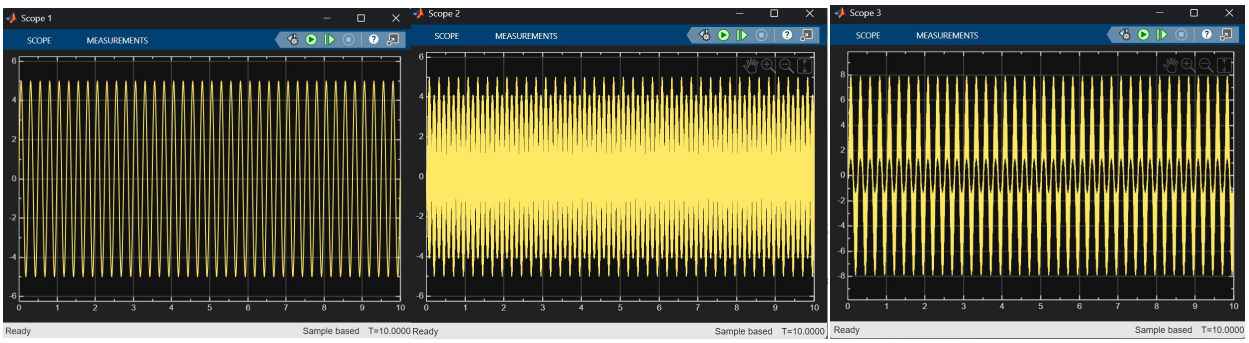
Matlab Simulink



Parameters



Output



Discussion: In this experiment, Single Side Band Suppressed Carrier (SSB-SC) modulation and demodulation were implemented using MATLAB. The results showed that unlike Double Side Band (DSB), SSB transmits only one sideband while suppressing the carrier and the other sideband, making it more bandwidth-efficient.The phase shift method was applied to generate and recover the signal. By introducing a 90° phase difference and combining signals properly, one sideband was canceled while the desired sideband remained. During demodulation, the product modulator and low-pass filter successfully recovered the original message signal.Overall, the experiment demonstrated that SSB modulation reduces bandwidth, improves efficiency, and provides clear signal recovery, though it requires accurate synchronization between transmitter and receiver.